



Significant Geomagnetic Storm Expected Beginning Today

Multiple solar disturbances are expected to reach Earth within a short period. Model guidance supports an initial arrival Friday, followed by additional disturbances that could strengthen or extend storm conditions through the weekend. Energetic-particle measurements from the ACE spacecraft were also rising and steepening by issuance. This is an early sign that the leading disturbance is approaching Earth. It increases confidence in arrival but does not indicate how strong the geomagnetic storm will become.

KEY POINTS

- There is a 65% chance that geomagnetic storming will begin between 1800 UTC Friday and 0000 UTC Saturday.
- The earliest possible onset is 1200 UTC Friday. The latest expected onset is 0600 UTC Saturday.
- There is an 85% chance of G3 or greater conditions, including a 5% chance of G5 conditions.
- Storm conditions may last 24-48 hours.
- Model runs 56381 and 56382 place the first arrivals near 2244 UTC and 2348 UTC Friday. Run 56385 adds another expected arrival near 0623 UTC Saturday. Run 56387 extends the sequence to near 0159 UTC Sunday. Together, the guidance indicates a series of disturbances rather than one short event.
- Later arrivals could strengthen the initial storm or produce renewed activity through the weekend.
- The ACE particle rise indicates that the first disturbance is approaching, but it cannot show the magnetic field carried by that disturbance.
- That magnetic field will be a major factor in determining storm strength and cannot be measured reliably until the disturbance reaches upstream spacecraft.

SPACE WEATHER RISK OUTLOOK

Sector	Day 1 Fri (May 10)	Day 2 Sat (May 11)	Day 3 Sun (May 12)
Power Grid	High	High	High
Aviation and Radiation	Moderate	High	Moderate
Satellite Operations	High	High	High
Communications and Navigation	High	High	High
Human Spaceflight	Moderate	High	Moderate
Emergency Management	Moderate	High	Moderate

These levels describe the recommended readiness posture. Actual effects will depend on storm strength, duration, location, and the vulnerability of individual systems.

Risk Levels:

Little to None

Minor

Moderate

High

Extreme

ACTIVE PRODUCTS

- Space Weather Warning: In effect for expected G3-G4 geomagnetic storm conditions through 0600 UTC Monday, May 13.
- Previous Space Weather Watch: Replaced by this Warning.
- Observed Space Weather Alert: Not issued for this event as of 1230 UTC Friday.
- Next update: No later than 0030 UTC Saturday, May 11, or sooner if conditions change.

POTENTIAL IMPACTS

- Power systems may experience voltage-control problems, transformer stress, and isolated equipment trips.
- Electric currents in the ground may affect power-grid equipment, pipeline monitoring, and corrosion-control systems.
- Increased atmospheric drag may lower satellite orbits and make tracking and orbit predictions less accurate.
- HF radio, satellite navigation, and some satellite communications may be disrupted, particularly at higher latitudes.
- Continued flare activity may cause additional radio blackouts on the sunlit side of Earth.
- Aurora may be visible across much of the United States and potentially farther south where skies are dark and clear.

WHAT TO DO NOW

- Put sector-specific response plans into effect where appropriate.
- Prepare for rapid changes after the first disturbance reaches Earth.
- Maintain heightened monitoring through the 0030 UTC Saturday update.
- Remain prepared for additional or renewed activity through Sunday rather than focusing on one predicted arrival.

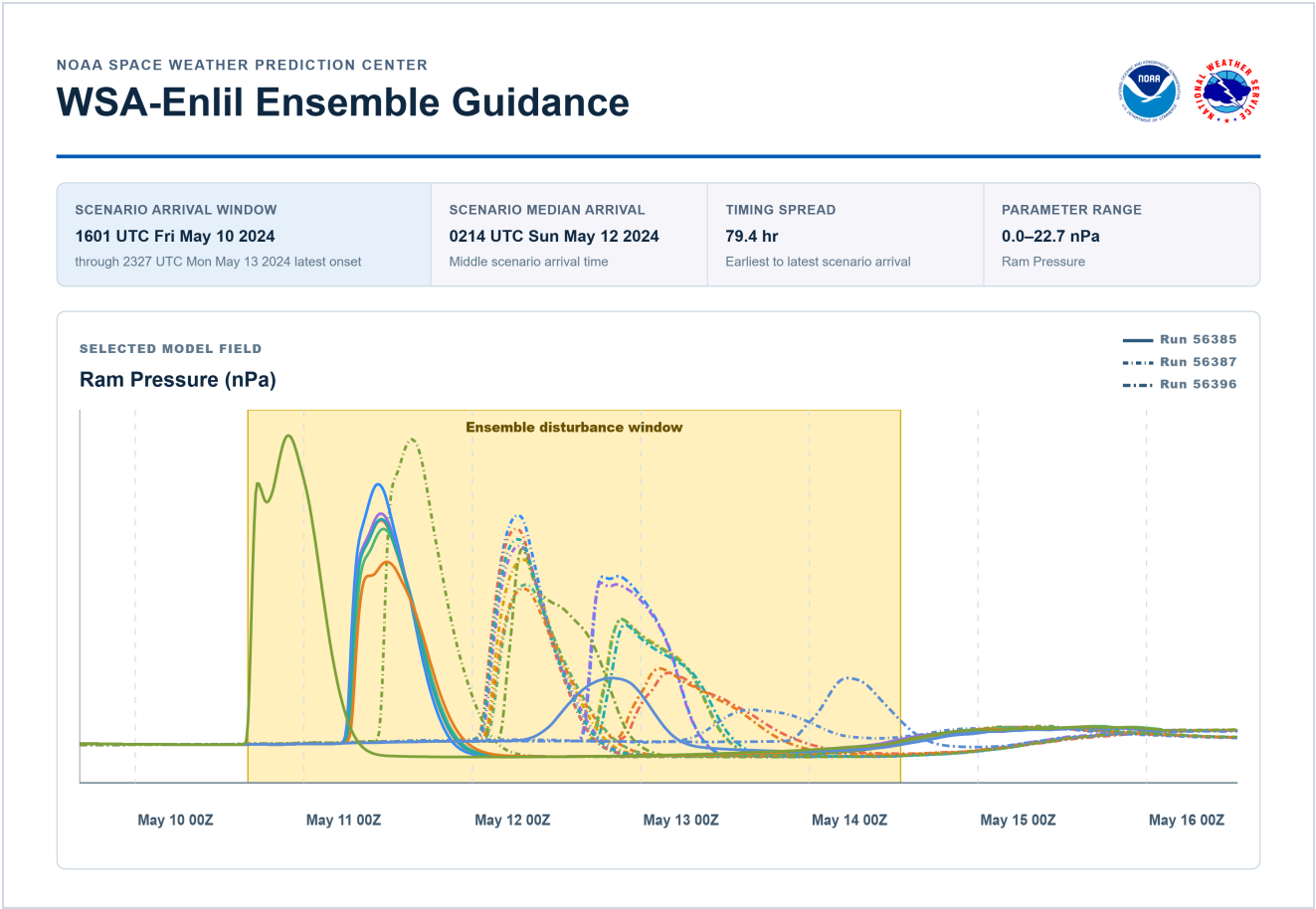
WHAT TO WATCH NEXT

- Whether the Saturday and Sunday disturbances strengthen or extend the initial storm.
- Observed alerts confirming G3, G4, or G5 conditions.
- Additional solar eruptions that could extend activity into Monday.

FOR MORE INFORMATION

For current forecasts, warnings, alerts, and space weather information, visit www.spaceweather.gov.
Questions or requests for decision support may be directed to the Space Weather Prediction Center at swpc.customersupport@noaa.gov or (303) 497-0016.
You are receiving this email because you are a valued space weather partner.

MEDIA



Initial Space Weather Warning graphic for the May 2024 geomagnetic storm event.